AUTOMATIC WATER IRRIGATION SYSTEM

Presented by: TEJASH SHUKLA

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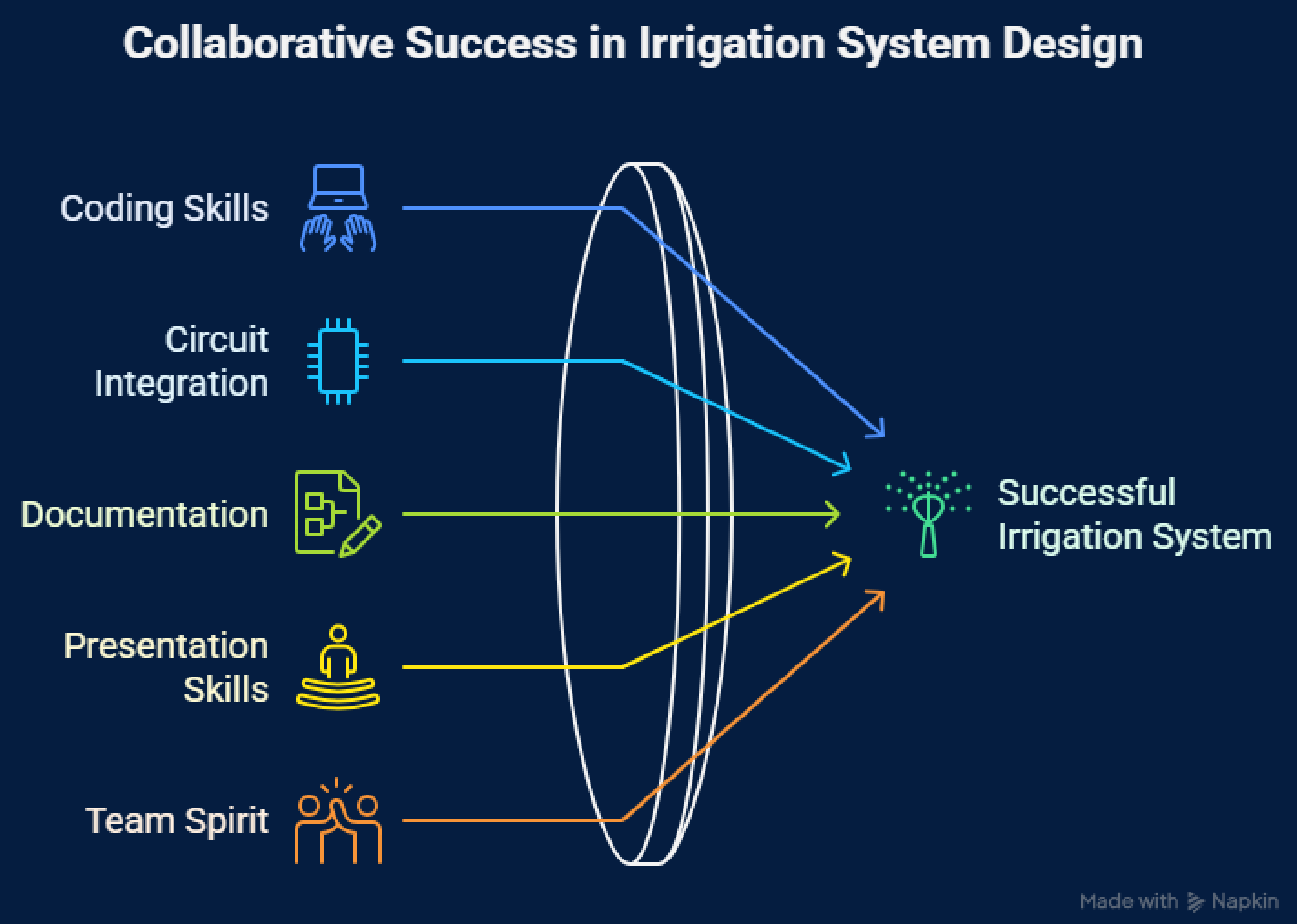
01 ABOUT US

Our team of eight dedicated members collaboratively developed the "Automatic water Irrigation System " Each member contributed to coding, circuit setup, research, documentation, and presentation, ensuring smooth project execution and success.

Name	**Role / Contribution**
Tejash Shukla	Developed the core Arduino code to run the project efficiently.
Savariya	Executed proper wiring and connection of all key electronic components.
Shivam Pandey	Conducted research on project requirements, working principles, and real-life applications.
Sakshi Parmar	Managed photography and video shooting for project documentation and demonstration.
Suraj Chaurasiya	Designed and created the PowerPoint presentation to showcase the project.
Shivani Meena	Assisted in preparing and formatting the PowerPoint presentation.
Sanskriti Shrivastava	Maintained and organized the project record file.
Stuti Sharma	Contributed to the preparation and presentation of the project record file.



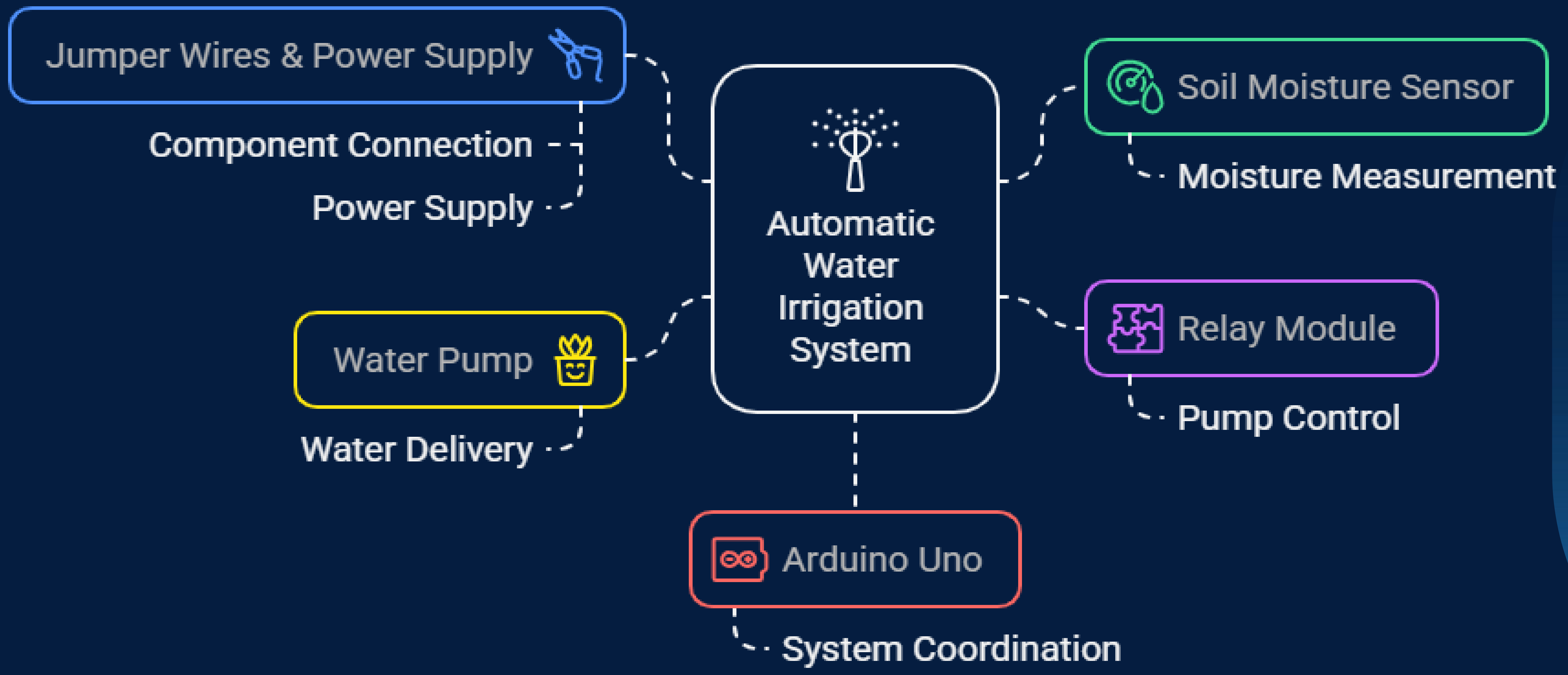
02 ABOUT PROJECT

Our project automates irrigation based on soil moisture, ensuring plants get the right amount of water, conserving water, reducing manual labor, and promoting healthy growth.

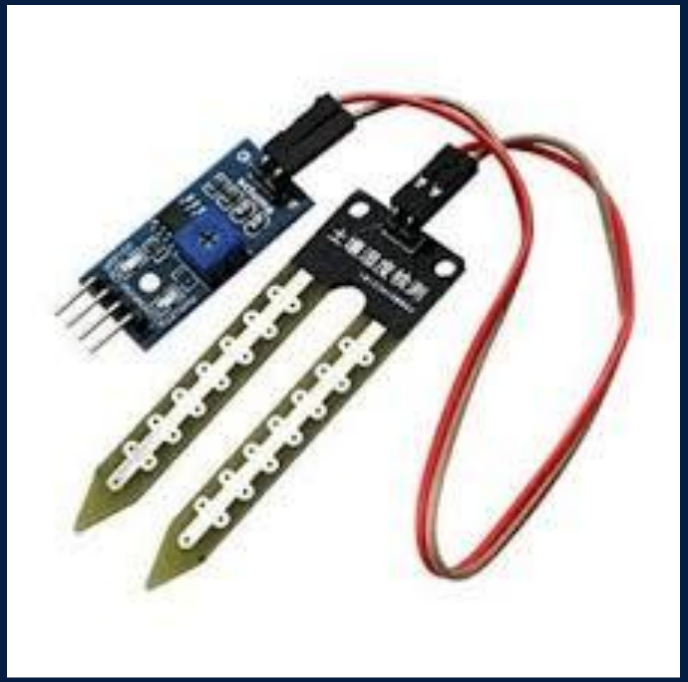
2. Objectives

- To monitor soil moisture continuously using a sensor.
- To automatically turn ON/OFF the water pump based on the moisture level.
- To ensure that plants get sufficient water without human intervention.
- To save water and improve agricultural efficiency.

03 Automatic Water Irrigation System Components



1. AURDINO UNO (R3)



2. MOISTURE SENSOR

3. RELAY MODULE

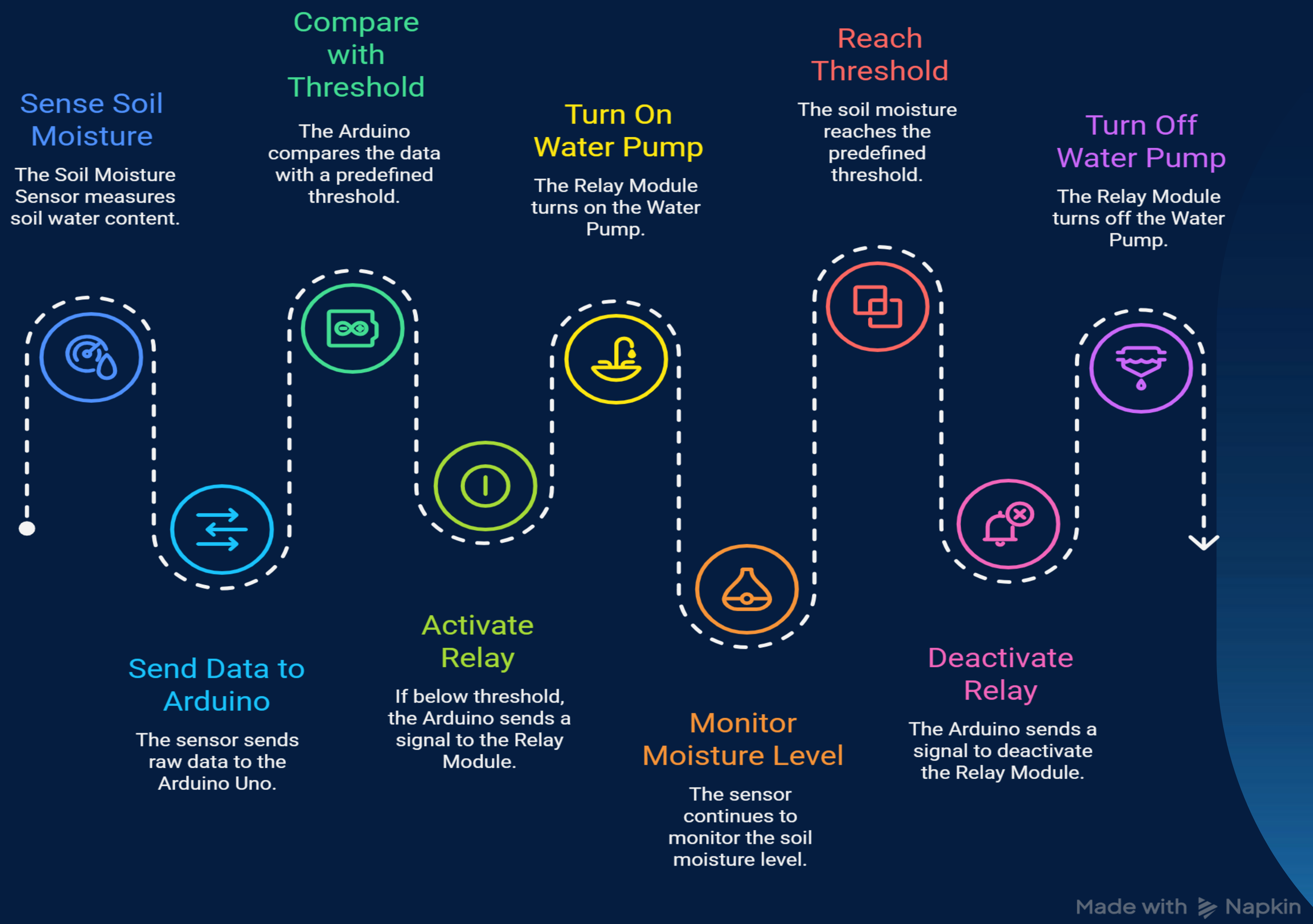
4. DC WATER PUMP

6. JUMPER WIRE

5. 9V BATTERY

04 WORKING

1. **Sensing the Need:** The **Soil Moisture Sensor**(connected to an Arduino Analog Pin) measures the soil's water content and sends the raw data (voltage) to the Arduino Uno.
2. **Decision Making (The Code):** The **Arduino Uno's code** constantly reads this value and compares it to a pre-defined **Threshold** (the point where the soil is considered 'dry' and needs water).
3. **Activating the Pump:** If the sensor value is below the threshold, the Arduino sends a **HIGH digital signal** to the **Relay Module** (connected to a Digital Pin). The Relay acts as an electrical switch, turning the high-power **Water Pump** ON.
4. **Automatic Shut-off:** The pump runs until the soil moisture sensor detects that the water level is **above the threshold**. The Arduino then sends a **LOW digital signal** to the Relay, turning the Water Pump OFF, thus completing the cycle.



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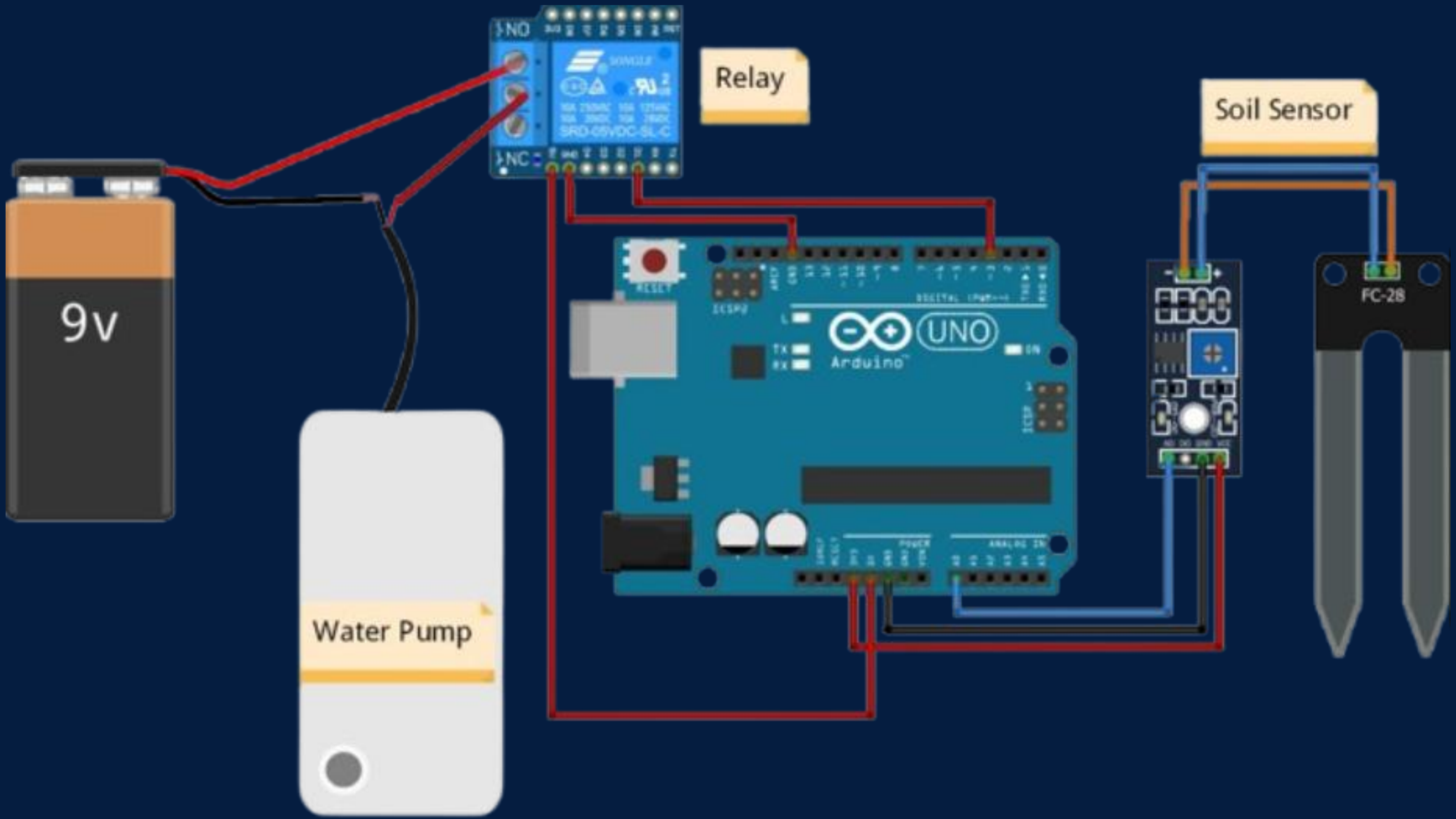
CONNECTIONS



06 SOURCE CODE

water_irrigation_system.ino

```
int sensor_pin = A0;
int output_value;
bool pumpOn = false;

int dryThreshold = 55;
int wetThreshold = 65;

void setup() {
  pinMode(3, OUTPUT);
  Serial.begin(9600);
  digitalWrite(3, HIGH); // Pump OFF
  Serial.println("WATER POURING!!");
  delay(2000);
}

void loop() {
  output_value = analogRead(sensor_pin);
  output_value = map(output_value, 900, 300, 0, 100);

  Serial.print("Moisture: ");
```

CODE

```
  Serial.print(output_value);
  Serial.println("%");

  if (!pumpOn && output_value < dryThreshold) {
    digitalWrite(3, LOW);
    pumpOn = true;
    Serial.println("Pump turned ON (Soil dry)");
    delay(5000); // let sensor stabilize
  }
  else if (pumpOn && output_value >= wetThreshold) {
    digitalWrite(3, HIGH);
    pumpOn = false;
    Serial.println("Pump turned OFF (Soil moist)");
    delay(2000); // prevent quick rechecking
  }

  delay(1000);
}
```

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MEMORIES DURING CONSTRUCTION

08 REAL LIFE APPLICATION

Smart Irrigation Systems

- **Precision Agriculture**
  - Optimized Crop Yields
  - Large-Scale Management
  - Resource Efficiency
  - Automated Nutrient Delivery
- **Landscaping & Commercial Property**
  - Water Conservation
  - Aesthetic Consistency
  - System Integrity
- **Specialized Horticulture**
  - Greenhouses & Nurseries
  - Vertical/Indoor Farms

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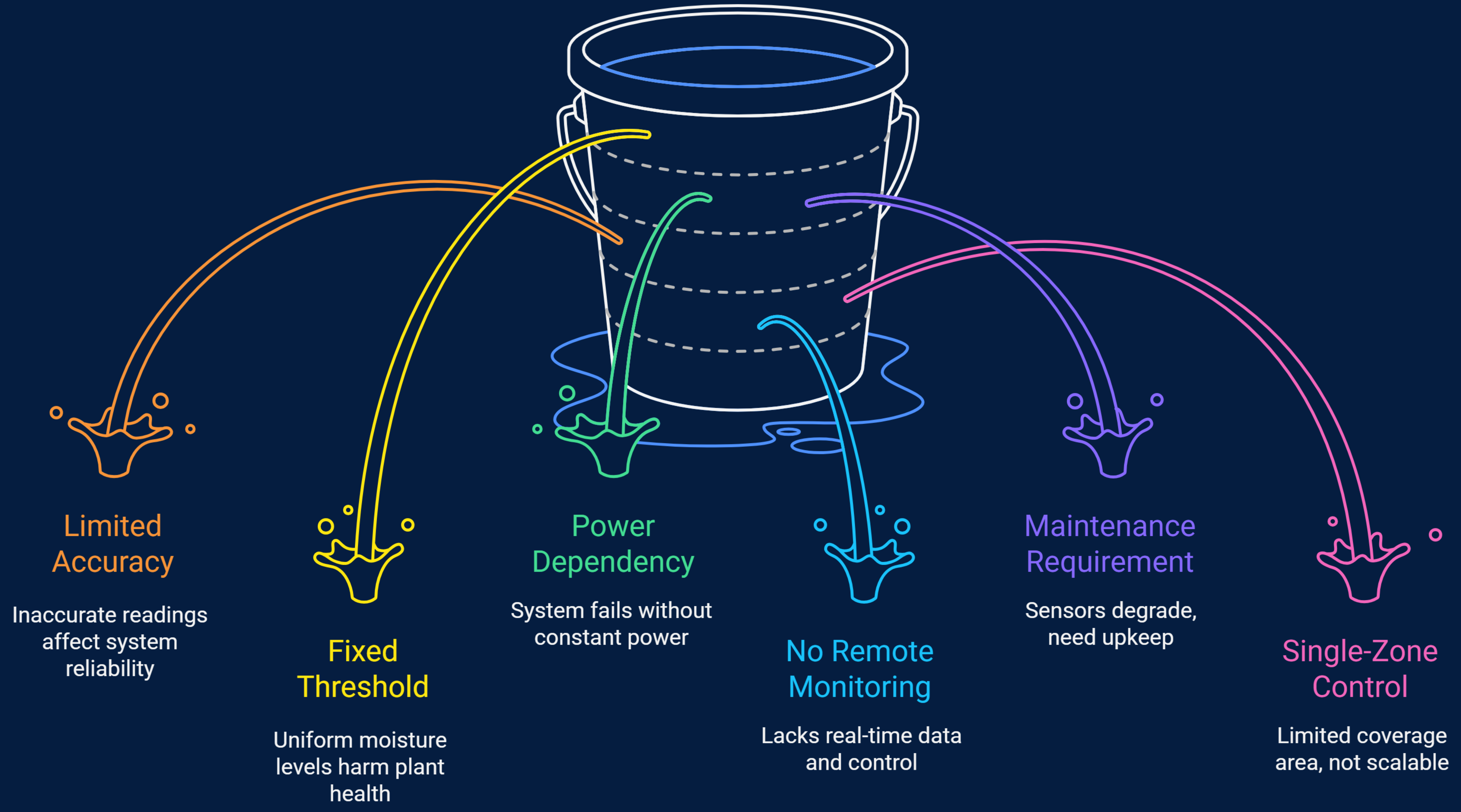
DRAWBACK'S



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Looking Ahead: What's Next for the Project?

Adaptive Thresholds

Adaptive thresholds are urgent for crop-specific moisture control.

IoT Integration

IoT integration is crucial for real-time data access.

Capacitive Sensors

Capacitive sensors offer stable readings but are less urgent.

Multi-Zone Control

Multi-zone control enhances irrigation efficiency but is less urgent.

REFERENCES & TOOLS

- Git hub Source Code 👉 [LINK HERE]
- WIKIPEDIA 👉 [LINK HERE]
- NAPKIN Ai (Tool) 👉 [LINK HERE]

THANK YOU

CONNECT WITH US.

ironmanb56@gmail.com